E.1 Generalizing prior

We define the following data generating process that constitutes a prior which generalizes to arbitrary, unseen symbols.

$$
\begin{aligned}
r &\sim \text{Bernoulli}\left(p = \frac{1}{2}\right) \\
\mathbf{s}_1 &\sim \mathcal{N}\left(0, \frac{1}{d}\right) \\
\mathbf{s}_2 &\sim \begin{cases} \delta(\mathbf{s}_1) & r = 1 \\ \mathcal{N}\left(0, \frac{1}{d}\right) & r = 0 \end{cases} \\
\mathbf{z}_1 &\sim \mathcal{N}\left(\mathbf{s}_1, \frac{\sigma^2}{d}\right) \\
\mathbf{z}_2 &\sim \mathcal{N}\left(\mathbf{s}_2, \frac{\sigma^2}{d}\right)
\end{aligned}
$$

The quantity r represents either a *same* or *different* relation. Variables $\mathbf{s}_1, \mathbf{s}_2$ are symbols matching their description in Section 2. The notation $\delta(\mathbf{s}_1)$ denotes a Delta distribution centered at $\mathbf{s}_1$. Hence, $\mathbf{s}_1 = \mathbf{s}_2$ if $r = 1$, and differ otherwise. Typically, we consider the noiseless case $\sigma^2 = 0$, but to develop a Bayesian treatment, we allow $\sigma^2 > 0$. We approximate the noiseless case by considering $\sigma^2 \to 0$.

The Bayes optimal classifier is

$$
\hat{y}_{bayes} = \begin{cases} 1 & p(r = 1 \,|\, \mathbf{z}_1, \mathbf{z}_2) \geq \frac{1}{2} \\ 0 & \text{otherwise} \end{cases} \tag{E.1}
$$

From Bayes rule, we know that

$$
p(r \,|\, \mathbf{z}_1, \mathbf{z}_2) \propto p(\mathbf{z}_1, \mathbf{z}_2 \,|\, r)\, p(r)\,.
$$

Since r is sampled with equal probability 1 or 0, we have simply

$$
p(r \,|\, \mathbf{z}_1, \mathbf{z}_2) \propto p(\mathbf{z}_1, \mathbf{z}_2 \,|\, r)\,.
$$

We use the notation $\mathcal{N}(\mathbf{x}; \mu, \sigma^2)$ to mean the PDF of a Normal distribution evaluated at $\mathbf{x}$, with mean μ and covariance $\sigma^2 \mathbf{I}$. We then compute

$$
\begin{aligned}
p(\mathbf{z}_1, \mathbf{z}_2 \,|\, r = 1) &= \int \mathcal{N}\left(\mathbf{z}_1; \mathbf{s}, \frac{\sigma^2}{d}\right) \mathcal{N}\left(\mathbf{z}_2; \mathbf{s}, \frac{\sigma^2}{d}\right) \mathcal{N}\left(\mathbf{s}; \mathbf{0}, \frac{\sigma^2}{d}\right) d\mathbf{s} \\
&= \left(\frac{d}{2\pi\sqrt{\sigma^2(2+\sigma^2)}}\right)^d \exp\left\{-\frac{d}{2\sigma^2}\left(\frac{1+\sigma^2}{2+\sigma^2}\left(||\mathbf{z}_1||^2 + ||\mathbf{z}_2||^2\right) - \frac{2}{2+\sigma^2}\left(\mathbf{z}_1 \cdot \mathbf{z}_2\right)\right)\right\}, \qquad \text{(E.2)} \\
p(\mathbf{z}_1, \mathbf{z}_2 \,|\, r = 0) &= \int\int \mathcal{N}\left(\mathbf{z}_1; \mathbf{s}_1, \frac{\sigma^2}{d}\right) \mathcal{N}\left(\mathbf{z}_2; \mathbf{s}_2, \frac{\sigma^2}{d}\right) \mathcal{N}\left(\mathbf{s}_1; \mathbf{0}, \frac{\sigma^2}{d}\right) \mathcal{N}\left(\mathbf{s}_2; \mathbf{0}, \frac{\sigma^2}{d}\right) d\mathbf{s}_1\, d\mathbf{s}_2 \\
&= \left(\frac{d}{2\pi(1+\sigma^2)}\right)^d \exp\left\{-\frac{d}{2}\left(\frac{1}{1+\sigma^2}\right)\left(||\mathbf{z}_1||^2 + ||\mathbf{z}_2||^2\right)\right\}. \qquad \text{(E.3)}
\end{aligned}
$$

Using Eq (E.2) and (E.3), we compute

$$
p(r = 1 \,|\, \mathbf{z}_1, \mathbf{z}_2) = \frac{p(\mathbf{z}_1, \mathbf{z}_2 \,|\, r = 1)}{p(\mathbf{z}_1, \mathbf{z}_2 \,|\, r = 1) + p(\mathbf{z}_1, \mathbf{z}_2 \,|\, r = 0)}\,,
$$

which we plug back into Eq (E.1) to obtain our Bayes classifier under a generalizing prior.

E.2 Memorizing prior

The data generating process for a model that memorizes the training data is similar to the generalizing model, but the crucial difference is that the symbols $\mathbf{s}$ are now distributed uniformly across the training symbols rather than sampled from their population distribution.

Let $\hat{\mathbf{s}}_1, \hat{\mathbf{s}}_2, \ldots, \hat{\mathbf{s}}_L$ be the set of L training symbols. Then the data generating process is given by

$$
\begin{aligned}
r &\sim \text{Bernoulli}\left(p = \frac{1}{2}\right) \\
\mathbf{s}_1 &\sim \text{Uniform}\left\{\hat{\mathbf{s}}_1, \hat{\mathbf{s}}_2, \ldots, \hat{\mathbf{s}}_L\right\} \\
\mathbf{s}_2 &\sim \begin{cases} \delta(\mathbf{s}_1) & r = 1 \\ \text{Uniform}\left\{\hat{\mathbf{s}}_1, \hat{\mathbf{s}}_2, \ldots, \hat{\mathbf{s}}_L\right\} \setminus \{\mathbf{s}_1\} & r = 0 \end{cases} \\
\mathbf{z}_1 &\sim \mathcal{N}\left(\mathbf{s}_1, \frac{\sigma^2}{d}\right) \\
\mathbf{z}_2 &\sim \mathcal{N}\left(\mathbf{s}_2, \frac{\sigma^2}{d}\right)
\end{aligned}
$$

As before, we compute the probabilities $p(\mathbf{z}_1, \mathbf{z}_2 \,|\, r = 1)$ and $p(\mathbf{z}_1, \mathbf{z}_2, \,|\, r = 0)$, which are given by

$$
\begin{aligned}
p(\mathbf{z}_1, \mathbf{z}_2 \,|\, r=1) &= \frac{1}{L}\sum_{i=1}^{L} p(\mathbf{z}_1, \mathbf{z}_2 \,|\, \hat{\mathbf{s}}_i) \\
&= \frac{1}{L}\sum_{i=1}^{L} \mathcal{N}\left(\mathbf{z}_1; \hat{\mathbf{s}}_i, \frac{\sigma^2}{d}\right) \mathcal{N}\left(\mathbf{z}_2; \hat{\mathbf{s}}_i, \frac{\sigma^2}{d}\right) \\
&= \left(\frac{d}{2\pi\sigma^2}\right)^d \exp\left\{-\frac{d}{2\sigma^2}\left(\frac{1}{2}\left(||\mathbf{z}_1||^2 + ||\mathbf{z}_2||\right)^2 - \mathbf{z}_1 \cdot \mathbf{z}_2\right)\right\} \left(\frac{1}{L}\sum_{i=1}^{L} \exp\left\{-\frac{d}{\sigma^2}\left|\left|\hat{\mathbf{s}}_i - \frac{\mathbf{z}_1 + \mathbf{z}_2}{2}\right|\right|^2\right\}\right), \qquad \text{(E.4)} \\
p(\mathbf{z}_1, \mathbf{z}_2 \,|\, r=0) &= \frac{1}{L(L-1)}\sum_{i \neq j} p(\mathbf{z}_1 \,|\, \hat{\mathbf{s}}_i)\, p(\mathbf{z}_2 \,|\, \hat{\mathbf{s}}_j) \\
&= \frac{1}{L(L-1)}\sum_{i \neq j}^{L} \mathcal{N}\left(\mathbf{z}_1; \hat{\mathbf{s}}_i, \frac{\sigma^2}{d}\right) \mathcal{N}\left(\mathbf{z}_2; \hat{\mathbf{s}}_j, \frac{\sigma^2}{d}\right) \\
&= \frac{1}{L(L-1)}\sum_{i \neq j} \left(\frac{d}{2\pi\sigma^2}\right)^d \exp\left\{-\frac{d}{2\sigma^2}\,||\mathbf{z}_1 - \hat{\mathbf{s}}_i||^2\right\} \exp\left\{-\frac{d}{2\sigma^2}\,\left|\left|\mathbf{z}_2 - \hat{\mathbf{s}}_j\right|\right|^2\right\}. \qquad \text{(E.5)}
\end{aligned}
$$

Using Eq (E.4) and (E.5), we compute Eq (E.1) to obtain our Bayes classifier under a memorizing prior.

F Rich and lazy scaling

We review rich and lazy regime scaling in our setting. In particular, we consider learning dynamics as we increase the input dimension d (Saad & Solla, 1995; Biehl & Schwarze, 1995; Goldt et al., 2019). This setting differs from other rich-regime studies, where scaling is considered with respect to increasing width m. In particular, *maximal update* (μP) and the related mean-field parameterizations consider an infinite-width limit (Yang & Hu, 2021; Mei et al., 2018; Rotskoff & Vanden-Eijnden, 2022). Our analysis holds m fixed.

Recall that our model is given by

$$
f(\mathbf{x}; \theta) = \frac{1}{\gamma\sqrt{d}}\sum_{i=1}^{m} a_i\, \phi(\mathbf{w}_i \cdot \mathbf{x}).
$$

Let $\theta(t)$ be the value of the parameters θ at time-step t. Crucially, to permit a valid interpolation between rich and lazy learning regimes, our MLP is *centered*: $f(\mathbf{x}; \theta(0)) = 0$. Following Chizat et al. (2019), we enforce centering by subtracting the initial logit from every prediction. Hence, our classifier takes the form

$$
\tilde{f}(\mathbf{x}; \theta) = f(\mathbf{x}; \theta) - f(\mathbf{x}; \theta(0)).
$$

We use $\tilde{f}$ as our centered MLP in all experiments.

To see how changing γ interpolates between rich and lazy learning regimes, recall that learning richness is a description of activation change over the course of training. One way to operationalize this description is to define *rich* learning as the case in which parameters θ change substantially in comparison with changes in the model output $\tilde{f}$, and *lazy* learning as the case in which θ change very little with respect to the model output.

Consider the change in $\tilde{f}$ after one step of gradient descent. For a learning rate α and training set size P, we update our parameters as

$$
\begin{aligned}
\theta(1) &= \theta(0) - \alpha \nabla_\theta \left(\frac{1}{P}\sum_{p=1}^{P} \mathcal{L}(y_p, \tilde{f}(\mathbf{x}_p; \theta))\right) \\
&= \theta(0) - \frac{\alpha}{P}\sum_{p=1}^{P} (y_p - \sigma(\tilde{f}(\mathbf{x}_p; \theta)) \nabla_\theta \tilde{f}(\mathbf{x}_p; \theta).
\end{aligned}
$$

Note that for an input $\mathbf{x}$,

$$
\begin{aligned}
\frac{\partial \tilde{f}}{\partial a_i} &= \frac{1}{\gamma\sqrt{d}}\phi(\mathbf{w}_i \cdot \mathbf{x}), \\
\frac{\partial \tilde{f}}{\partial \mathbf{w}_i} &= \frac{1}{\gamma\sqrt{d}} a_i\, \phi'(\mathbf{w}_i \cdot \mathbf{x})\, \mathbf{x}.
\end{aligned}
$$

Define

$$
\begin{aligned}
\Delta a_i &\equiv \frac{1}{P}\sum_{p=1}^{P} (y_p - \sigma(\tilde{f}(\mathbf{x}_p; \theta))) \frac{\partial \tilde{f}}{\partial a_i}, \\
\Delta \mathbf{w}_i &\equiv \frac{1}{P}\sum_{p=1}^{P} (y_p - \sigma(\tilde{f}(\mathbf{x}_p; \theta))) \frac{\partial \tilde{f}}{\partial \mathbf{w}_i}.
\end{aligned}
$$

Substituting our weight updates into our model reveals that

$$
\begin{aligned}
\tilde{f}(\mathbf{x}; \theta(1)) = \frac{\alpha}{\gamma^2 d}\sum_{i=1}^{m} \Big[&\Delta a_i\, \phi(\Delta w_i \cdot \mathbf{x})/(\gamma\sqrt{d}) \\
&+ \Delta a_i\, \phi(\mathbf{w}(0) \cdot \mathbf{x}) \\
&+ a_i(0)\, \phi(\Delta \mathbf{w}_i \cdot \mathbf{x}) \Big]. \qquad \text{(F.1)}
\end{aligned}
$$

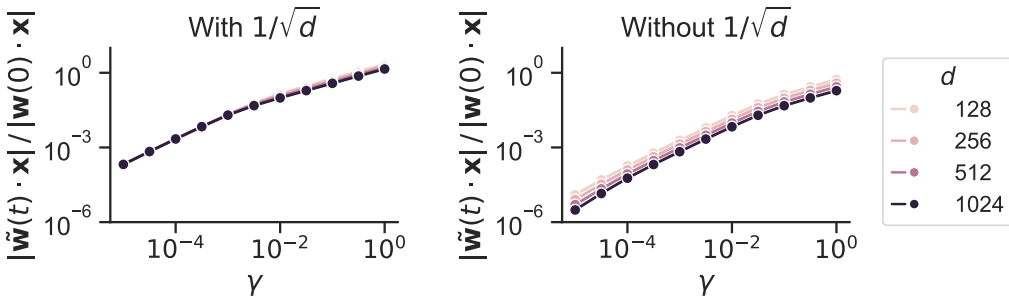

Figure F1: **Activation scales with γ and d.** We plot the average absolute activation change across 6000 test examples as a function of γ (for $m = 4096$), normalized by the initial activation size $|\mathbf{w}(0)\cdot\mathbf{x}|$. Higher γ leads to more activation change. In the absence of a $1/\sqrt{d}$ prefactor, the activation change scales inversely with d. Including the $1/\sqrt{d}$ prefactor suppresses this change.

Observe that $|\Delta a_i \phi(\mathbf{w}(0)\cdot\mathbf{x})| = \mathcal{O}_d(1)$ and $|a_i(0)\phi(\Delta\mathbf{w}_i\cdot\mathbf{x})| = \mathcal{O}_d(1/\sqrt{d})$ for a test point $\mathbf{x}$. If we adopt a learning rate $\alpha = \gamma^2 d$, then we have overall

$$|\tilde{f}(\mathbf{x};\theta(1))| = \mathcal{O}_d(1)\,. \tag{F.2}$$

In this way, we find that the model output changes by a constant amount relative to the input dimension d. Meanwhile, the parameters change by a total magnitude

$$||\theta(1)-\theta(0)|| = \mathcal{O}_d(\alpha(|\Delta a_i| + ||\Delta\mathbf{w}_i||) = \mathcal{O}_d(\gamma\sqrt{d})\,.$$

At initialization, we have that

$$||\theta(0)|| = \mathcal{O}_d(|a_i(0)| + ||\mathbf{w}_i(0)||) = \mathcal{O}_d(\sqrt{d})\,,$$

so the change in weights relative to the scale of their initialization is simply

$$||\theta(1)-\theta(0)||/||\theta(0)|| = \mathcal{O}_d(\gamma)\,. \tag{F.3}$$

All together, after one gradient step, while the model output changes by a constant amount with respect to d, the model parameters change by γ relative to their initialization. For $\gamma \to 0$, the initialization dominates (even as the model output changes), resulting in lazy learning. For increasing γ, the parameters move proportionally further from their initialization, resulting in progressively rich learning.

Peculiar to our setting is the additional $1/\sqrt{d}$ factor in the output scale of the MLP, not found in other rich-regime studies that consider width scaling (Yang & Hu, 2021; Mei et al., 2018; Rotskoff & Vanden-Eijnden, 2022). In the absence of the $1/\sqrt{d}$ factor, Eqs (F.1) and (F.2) suggest that we should adjust our learning rate to be $\alpha = \gamma^2$ in order to maintain a stable $\mathcal{O}_d(1)$ output change with increasing d. However, the relative weight change in Eq (F.3) now becomes $\mathcal{O}_d(\gamma/\sqrt{d})$. For fixed γ, a model becomes lazier as d increases. Hence, to maintain consistent richness, we require an additional $1/\sqrt{d}$ prefactor on the MLP (along with the corresponding $\alpha = \gamma^2 d$ learning rate).

Figure F1 illustrates these conclusions. Increasing γ increases the change in activations $|\tilde{\mathbf{w}}(t)\cdot\mathbf{x}|$ for a test example $\mathbf{x}$. In the absence of the $1/\sqrt{d}$ prefactor, increasing d also decreases the change in activations.

G Model and task details

We enumerate all model and task configurations in this Appendix. Exact details are available in our code, https://github.com/wtong98/equality-reasoning, which can be run to reproduce all plots in this manuscript.

G.1 Model

In all experiments, we use a two-layer MLP without biases that takes inputs $\mathbf{x} \in \mathbb{R}^d$ and outputs

$$f(\mathbf{x}) = \frac{1}{\gamma\sqrt{d}}\sum_{i=1}^{m} a_i \phi(\mathbf{w}_i\cdot\mathbf{x})\,,$$

where ϕ is a point-wise ReLU nonlinearity and γ is a hyperparamter that governs learning richness. Our MLP is centered using the procedure described in Appendix F. To produce a classification, f is passed through a standard logit link function

$$\hat{y} = \frac{1}{1+e^{-f}}\,.$$

Parameters are initialized based on μP (Yang et al., 2022). Specifically, we initialize our weights as

$$a_i \sim \mathcal{N}(0, 1/m)\,,$$
$$\mathbf{w}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}/m)\,.$$

We train the model using stochastic gradient descent on binary cross entropy loss. Following Atanasov et al. (2024), we set the learning rate α as $\alpha = \gamma^2 d\,\alpha_0$ for $\gamma \le 1$ and $\alpha = \gamma\sqrt{d}\,\alpha_0$ for $\gamma > 1$. The base learning rate α_0 is task-specific, and varies from 0.01 to 0.5. To measure a model's performance, we train for a large, fixed number of iterations past convergence in training accuracy, and select the best test accuracy from the model's history.

G.2 Same-Different

The same-different task consists of input pairs $\mathbf{z}_1, \mathbf{z}_2 \in \mathbb{R}^d$, where $\mathbf{z}_i = \mathbf{s}_i + \eta_i$. The labeling function y is given by

$$y(\mathbf{z}_1, \mathbf{z}_2) = \begin{cases} 1 & \mathbf{s}_1 = \mathbf{s}_2 \\ 0 & \mathbf{s}_1 \neq \mathbf{s}_2 \end{cases}\,.$$